

Daily Logic Drill #6 — Answers & Investigations

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Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

Answers and Methods.

Section A Rapid Recognition

1. I. “2 is prime.” II. “2 is odd.” Which are true?

Answer: I only.

Method: 2 is prime (I true). 2 is even, not odd (II false).

Investigate further: 2’s exceptional evenness (Day 1, B9) will keep showing up as the reason a statement about “all primes” needs independent per-statement checking rather than a blanket assumption.

2. I. “4 is a perfect square.” II. “4 is prime.” Which are true?

Answer: I only.

Method: $4 = 2^2$ (I true). $4 = 2 \times 2$ is composite, not prime (II false).

Investigate further: Evaluating I and II here requires no shared reasoning at all — they’re entirely independent facts about the same number.

3. I. “9 is odd.” II. “9 is prime.” Which are true?

Answer: I only.

Method: 9 is odd (I true). $9 = 3 \times 3$ is composite (II false).

Investigate further: This is Day 4’s A10 counterexample, now split into its two separate true/false components.

4. I. “Every square is a rectangle.” II. “Every rectangle is a square.” Which are true?

Answer: I only.

Method: I is true (Day 1, A3). II is false, e.g. a 2×3 rectangle (Day 1, A4).

Investigate further: This restates Day 1’s opening pair as an explicit I/II evaluation — the format is new, the content is a full sheet old.

5. I. “6 is a multiple of 2.” II. “6 is a multiple of 3.” Which are true?

Answer: Both.

Method: $6 = 2 \times 3$, so 6 is a multiple of both 2 and 3.

Investigate further: When both statements are true, the answer is simply “both” — don’t overthink combinations that don’t require it.

6. I. “5 is even.” II. “5 is prime.” Which are true?

Answer: II only.

Method: 5 is odd, not even (I false). 5 has exactly two divisors, 1 and 5 (II true).

Investigate further: The mirror image of A1: here it’s the primality statement that survives, not the parity one.

7. I. “0 is positive.” II. “0 is negative.” Which are true?

Answer: Neither.

Method: 0 is neither positive nor negative by definition.

Investigate further: 0 is a recurring “breaks the pattern” value across this pillar (Day 4’s B5, B7) — here it breaks a positive/negative dichotomy that every other real number satisfies.

8. True or false: “If both statements I and II in a pair are false, the correct choice is ‘neither’, not ‘I and II’.”

Answer: True.

Method: “I and II” means both are true; when both are false, the correct description is “neither” — these are opposite answer choices, never interchangeable.

Investigate further: A7 is exactly this situation in practice: both I and II false, so “neither” is the only correct choice, not a careless “I and II.”

9. I. “ $n = 2$ is a counterexample to ‘all primes are odd’.” II. “ $n = 9$ is a counterexample to ‘all odd numbers are prime’.” Which are true?

Answer: Both.

Method: I is true: 2 is prime but even (Day 4, A1). II is true: 9 is odd but composite (Day 4, A10).

Investigate further: This format — checking whether a proposed instance genuinely is a counterexample — is exactly Day 4’s skill, now folded into the I/II evaluation habit.

10. I. “1 is a prime number.” II. “1 is a perfect square.” Which are true?

Answer: II only.

Method: 1 is excluded from the primes by definition (a prime has exactly two distinct positive divisors; 1 has only one) (I false). $1 = 1^2$ (II true).

Investigate further: 1’s exclusion from primality (Day 4, B6) and its inclusion among perfect squares are both definitional facts worth having instantly available, not re-derived each time.

Section B Manipulation Drills

1. I. “ $n = 4$ is a multiple of 4.” II. “ $n = 4$ is a multiple of 8.” III. “ $n = 4$ is a multiple of 2.” Which are true?

Answer: I and III only.

Method: $4 = 4 \times 1$ (I true). 4 is not a multiple of 8 (II false). $4 = 2 \times 2$ (III true).

Investigate further: This is exactly the counterexample instance from Day 1's A9/Day 4's A5, now split into three independently-checked statements.

2. I. " n even $\Rightarrow n^2$ even" is true. II. Its converse is also true. III. Together, I and II mean " n even $\Leftrightarrow n^2$ even." Which are true?

Answer: All three.

Method: I is Day 1's B2. II is Day 1's B5-via-A5 (the converse " n^2 even $\Rightarrow n$ even" is also true). III is exactly Day 2's B4 classification (necessary and sufficient) restated as "iff."

Investigate further: This is a full recap of one entire thread running through Days 1–2, now confirmed as a single consistent package.

3. I. " $\sqrt{2}$ is irrational." II. " $\sqrt{2} + \sqrt{2}$ is irrational." III. " $\sqrt{2} - \sqrt{2}$ is irrational." Which are true?

Answer: I and II only.

Method: I is the classical fact (Day 5, D2). II: $\sqrt{2} + \sqrt{2} = 2\sqrt{2}$, still irrational (a nonzero rational multiple of an irrational is irrational). III: $\sqrt{2} - \sqrt{2} = 0$, which is rational, so III is false.

Investigate further: III is exactly Day 4's B2 cancellation trick in reverse — the same construction that disproves "irrational + irrational is always irrational" shows up here as a deliberately-included false option.

4. I. "Every multiple of 9 is a multiple of 3." II. "Every multiple of 3 is a multiple of 9." III. "Multiple of 3' is necessary for 'multiple of 9'." Which are true?

Answer: I and III only.

Method: I is true ($9 = 3 \times 3$). II is false ($n = 3$ is a multiple of 3 but not 9). III restates I in necessary/sufficient language (Day 2): "multiple of 9" $\Rightarrow$ "multiple of 3" is exactly I, so III is true.

Investigate further: I and III are the same fact in two vocabularies (Day 1's implication language and Day 2's necessary/sufficient language) — recognising that is itself part of the test.

5. I. " $2^{11} - 1$ is prime." II. "11 is prime." III. "If n is prime, $2^n - 1$ is prime." Which are true?

Answer: II only.

Method: I is false: $2^{11} - 1 = 2047 = 23 \times 89$ (Day 4, C4). II is true. III is false, disproved by exactly this instance ($n = 11$ prime, $2^{11} - 1$ not prime).

Investigate further: I is literally the counterexample that disproves III — when a statement provides its own disproof of another statement in the same list, use that connection rather than re-deriving from scratch.

6. I. " $4! + 1$ is prime." II. " $5! + 1$ is prime." III. " $n! + 1$ is prime for all n ." Which are true?

Answer: None of them.

Method: $4! + 1 = 25 = 5^2$, not prime (I false, Day 4 A8). $5! + 1 = 121 = 11^2$, not prime (II false). III is

false, already disproved at $n = 4$.

Investigate further: A “none of the above” answer is just as legitimate as any other combination — don’t assume at least one statement must be true.

7. I. “41 is prime.” II. “ $n = 41$ is a counterexample to ‘ $n^2 + n + 41$ is always prime’.” III. “ $n = 40$ is the smallest counterexample to ‘ $n^2 + n + 41$ is always prime’.” Which are true?

Answer: All three.

Method: 41 is prime (I true). $n = 41$ gives 41×43 , composite (II true, Day 4 B1). $n = 40$ gives $1681 = 41^2$, the smallest failure (III true, Day 4 C1).

Investigate further: Statements about the same famous polynomial, checked at three different levels of precision (a background fact, a valid-but-not-minimal counterexample, and the genuinely smallest one) — all can be true simultaneously without contradiction.

8. I. “The negation of ‘for all x , $P(x)$ ’ is ‘for all x , not $P(x)$ ’.” II. “The negation of ‘for all x , $P(x)$ ’ is ‘there exists x , not $P(x)$ ’.” III. “The negation of ‘there exists x , $P(x)$ ’ is ‘for all x , not $P(x)$ ’.” Which are true?

Answer: II and III only.

Method: I is false — negating a $\forall$ gives an $\exists$, not another $\forall$ (Day 3, A5). II is the correct rule (Day 3, A5/B2). III is the correct rule for the other direction (Day 3, A6/C4).

Investigate further: I is a plausible-sounding but wrong “fix” of II — exactly the kind of near-miss distractor a rushed reader mistakes for correct.

9. I. “The converse of a true statement is always true.” II. “The contrapositive of a true statement is always true.” III. “The negation of a true statement is always true.” Which are true?

Answer: II only.

Method: I is false (Day 1, A10). II is true (Day 3, A9). III is false — the negation of a true statement is always *false* (Day 3, A7), the opposite of what III claims.

Investigate further: III is a deliberate near-miss of a real fact (Day 3’s A7): swapping “false” for “true” turns a correct statement into an incorrect one, exactly the kind of misread this format is built to catch.

10. I. “Affirming the consequent is a valid rule.” II. “Denying the antecedent is a valid rule.” III. “Modus ponens is a valid rule.” Which are true?

Answer: III only.

Method: I and II are both named fallacies, not valid rules (Day 5, A1/A3). III is the basic valid inference rule (Day 5, A4).

Investigate further: Two of these three names sound equally formal and rule-like as the third — naming conventions alone don’t tell you validity; you have to know which is which.

Section C Substitution & Structure

1. For a positive integer n : I. “ n divisible by 4” is necessary for “ n divisible by 12.” II. “ n divisible by 4” is sufficient for “ n divisible by 12.” III. “ n divisible by 4” is necessary and sufficient for

“ n divisible by 8.” Which of I, II, III are true?

Answer: A) I only.

Method: I is true: $12 = 4 \times 3$, so $12 \mid n \Rightarrow 4 \mid n$ (Day 2, C6). II is false: $n = 4$ is divisible by 4 but not 12. III is false: $4 \mid n$ is necessary for $8 \mid n$ (since $8 = 4 \times 2$) but not sufficient ($n = 4$ counterexample again), so it's not both.

Investigate further: The same value $n = 4$ serves as the counterexample for both II and (half of) III — a single well-chosen instance can carry a lot of weight across a whole question.

2. I. “There exists a real x with $x^2 < 0$.” II. “For all reals x , $x^2 \geq 0$.” III. “The negation of II is I.” Which are true?

Answer: II and III only.

Method: I is false — no real number squares to a negative value. II is the true universal fact. III is true: negating “for all x , $x^2 \geq 0$ ” gives “there exists x , $x^2 < 0$ ” (Day 3’s quantifier-negation rule), which is exactly I.

Investigate further: I being false and III being true are fully compatible — III only claims I is the *correct negation* of II, not that I itself is true; a correct negation of a true statement is expected to be false (Day 3, A7).

3. Recall the flawed argument “ n^2 multiple of 4, therefore n multiple of 4” (Day 5, B1). I. “This argument commits the fallacy of affirming the consequent.” II. “ $n = 6$ is a valid counterexample to the underlying claim.” III. “The converse of the underlying claim (n mult. of 4 $\Rightarrow n^2$ mult. of 4) is what’s actually true.” Which are true?

Answer: All three.

Method: I is true (Day 5, B1’s diagnosis). II is true: $n = 6$ gives $n^2 = 36$ (mult. of 4) but 6 is not. III is true: “ n mult. 4 $\Rightarrow n^2$ mult. 4” is the actual valid direction (Day 1, C5-style substitution), and the flawed argument mistakes its converse for a valid inference.

Investigate further: This question makes explicit what “affirming the consequent” means concretely: treating a true statement’s converse as if it were the (also true) original.

4. I. “ $n = 1$ is a counterexample to ‘every positive integer has a prime factor’.” II. “ $n = 1$ is a counterexample to ‘every integer greater than 1 has a prime factor’.” III. “1 is neither prime nor composite.” Which are true?

Answer: I and III only.

Method: I is true (Day 4, B6): 1 is a positive integer with no prime factor. II is false: $n = 1$ does not satisfy the hypothesis “greater than 1” at all, so it cannot be a counterexample to that (different, and true) claim — a counterexample must make the hypothesis true and the conclusion false. III is the standard definitional fact.

Investigate further: II is a trap: reusing a familiar-looking value without checking it actually satisfies the hypothesis of *this* particular claim is a common way to wrongly call something a counterexample.

5. I. “The statement ‘if $n^2 - n$ is even, then n is even’ is false (Day 1, D5).” II. “ $n^2 - n$ is even for every integer n .” III. “ $n(n - 1)$ is even for every integer n .” Which are true?

Answer: All three.

Method: I is true (Day 1, D5, witnessed by $n = 3$). II is true: $n^2 - n = n(n - 1)$, a product of consecutive integers, always even. III is the same fact as II, merely written in factored form rather than expanded form.

Investigate further: II and III are not two independent facts — they’re the identical claim under two different (but equal) algebraic expressions, worth recognising rather than re-deriving separately.

6. I. “Every quadratic with integer coefficients that has an integer root has two integer roots.” II. “ $2x^2 - 3x + 1$ is a counterexample to I.” III. “Monic quadratics (leading coefficient 1) are never counterexamples to I.” Which are true?

Answer: II and III only.

Method: I is the false universal claim itself (false). II is true: $2x^2 - 3x + 1 = (2x - 1)(x - 1)$ has roots 1 and $\frac{1}{2}$ (Day 4, B4). III is true: for monic quadratics, an integer root forces the other root to also be an integer via Vieta’s formulas (Day 4, B4’s

Investigate further:

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Investigate further: I being the claim under test, and therefore false, doesn’t stop II and III (statements *about* I) from both being true — keep track of what each statement is actually asserting.

7. I. “ $3^5 + 2$ is composite (Day 4, C6).” II. “ $3^5 + 2 = 245$.” III. “ $245 = 5 \times 7^2$.” Which are true?

Answer: All three.

Method: $3^5 = 243$, so $3^5 + 2 = 245$ (II true). $245 = 5 \times 49 = 5 \times 7^2$ (III true), confirming 245 is composite (I true).

Investigate further: This question is a pure arithmetic-verification chain — each statement builds directly on confirming the previous one, the same discipline this pillar’s verify scripts apply computationally.

8. For real x : I. “ $x > 0$ ” is sufficient for “ $x^2 > 0$.” II. “ $x > 0$ ” is necessary for “ $x^2 > 0$.” III. “ $x \neq 0$ ” is necessary and sufficient for “ $x^2 > 0$.”

Answer: C) I and III only.

Method: I is true (Day 2, D1). II is false: $x = -1$ gives $x^2 = 1 > 0$ without $x > 0$ (Day 2, D1). III is true: $x \neq 0 \Rightarrow x^2 > 0$ (sufficient), and $x^2 > 0 \Rightarrow x \neq 0$ (necessary, since $x = 0$ gives $x^2 = 0$) — both directions hold.

Investigate further: I and III together show “ $x \neq 0$ ” is the tight (necessary-and-sufficient) characterisation, while “ $x > 0$ ” alone is only ever half the story — exactly the “weakest sufficient condition” idea from Day 2, B9.

Section D Challenge

1. For a positive integer n : I. “ n divisible by 6” is necessary and sufficient for “ n divisible by 2 and by 3.” II. “ n^2 divisible by 9” is necessary and sufficient for “ n divisible by 3.” III. “ n^2 divisible by 4” is necessary and sufficient for “ n divisible by 4.” Which of I, II, III are true?

Answer: D) I and II only.

Method: I is true (Day 2, B2/C1). II is true: 3 is prime, so $9 \mid n^2 \Rightarrow 3 \mid n^2 \Rightarrow 3 \mid n$ (necessary, Day 1 C1/Day 3 C8), and $3 \mid n \Rightarrow n^2 = 9k^2$ (sufficient, Day 1 C1) — both directions hold. III is false: $4 \mid n^2$ does not force $4 \mid n$, witnessed by $n = 6$ (the recurring counterexample from Day 1 C8, Day 4, Day 5).

Investigate further: The difference between II and III is exactly the difference between a prime modulus (3) and a prime-squared modulus ($4 = 2^2$) — the entire reason Day 1's C1/C8 pair existed was to seed this exact distinction for today.

2. Three people, X , Y , Z , each make one statement. X says: “ Y is lying.” Y says: “ Z is lying.” Z says: “ X and Y are both lying.” Given that exactly one of them is telling the truth, determine who.

Answer: Y is the truth-teller; X and Z are lying.

Method: Suppose X is the sole truth-teller: then “ Y is lying” is true, so Y lies, meaning Y 's statement “ Z is lying” is false, so Z tells the truth — but then both X and Z are truthful, contradicting “exactly one.” Suppose Z is the sole truth-teller: then “ X and Y are both lying” is true, so X lies, meaning X 's statement “ Y is lying” is false, so Y tells the truth — but then both Z and Y are truthful, again a contradiction. Suppose Y is the sole truth-teller: then Z lies (Y 's statement true), and X lies (since only Y is truthful), so X 's statement “ Y is lying” is false, correctly meaning Y is not lying — consistent. Z 's statement “ X and Y are both lying” is false (since Y isn't lying), correctly making Z a liar — consistent. This is the only self-consistent case.

Investigate further: This is exactly Day 3's D3 truth-teller puzzle scaled up to three people — the method (assume each person is the exceptional one in turn, check for self-contradiction) is identical, just with more cases to rule out.

3. Recall the flawed conclusion from Day 5, D2 (a student proves $\sqrt{2}$ irrational by contradiction, then writes “therefore $\sqrt{2}$ is rational”). I. “The algebraic derivation, up to the final sentence, is entirely correct.” II. “The final English conclusion correctly follows from the derived contradiction.” III. “A correctly-stated conclusion would be ‘ $\sqrt{2}$ is irrational’.” Which are true?

Answer: I and III only.

Method: I is true — every algebraic step (deriving p, q both even) is valid (Day 5, D2). II is false — the student's final sentence states the opposite of what the contradiction establishes. III is true — a contradiction derived from assuming rationality correctly proves irrationality.

Investigate further: This question makes Day 5's D2 error explicit by isolating exactly which part (the algebra) was fine and which part (the one-sentence conclusion) was not.

4. Let S be the statement: “If n is a multiple of 12, then n is a multiple of 3 and a multiple of 4.” I. S is true. II. S 's converse is true. III. “Multiple of 12” is necessary and sufficient for “multiple of 6.” Which are true?

Answer: I and II only.

Method: I is true: $12 = 3 \times 4$, so $12 \mid n \Rightarrow 3 \mid n$ and $4 \mid n$. II (the converse, “multiple of 3 and of 4 $\Rightarrow$ multiple of 12”) is true: $\gcd(3, 4) = 1$, so $\text{lcm}(3, 4) = 12$. III is false: “multiple of 12” is sufficient for “multiple of 6” ($12 = 6 \times 2$) but not necessary — $n = 6$ is a multiple of 6 but not of 12.

Investigate further: I and II together establish S is necessary and sufficient for “multiple of 3 and of 4” (Day 2’s iff) — III then tests whether that same strength of relationship carries over to a *different* target (“multiple of 6”), and it doesn’t.

5. Statement T : “If $2^n - 1$ is prime, then n is prime” (Day 4, D5). I. T ’s contrapositive is “if n is not prime, then $2^n - 1$ is not prime.” II. T ’s converse is false, witnessed by $n = 11$. III. “ $2^n - 1$ is prime” is necessary and sufficient for “ n is prime.” Which are true?

Answer: I and II only.

Method: I is true — negate and swap both clauses of T to get exactly this (Day 3’s construction rule). II is true: T ’s converse is “if n prime then $2^n - 1$ prime,” and $n = 11$ is prime with $2^{11} - 1 = 2047 = 23 \times 89$ composite (Day 4, C4), a genuine counterexample. III is false: T itself gives sufficiency, but the false converse (II) means necessity fails, so it’s sufficient but not necessary, not both (Day 4, D5’s exact classification).

Investigate further: This closing question is the pillar’s own toolkit turned on itself: constructing a contrapositive (Day 3), witnessing a false converse (Day 1/Day 4), and classifying necessity/sufficiency (Day 2) all applied to one statement, in one place, to close the multi-statement-evaluation day.